

From Risk Factors to Diagnosis : Predicting Cervical Biopsy Outcome with Supervised Machine Learning

Supervised learning on the UCI Cervical Cancer (Risk Factors) dataset

UM5BM753 Project - M2 AIDA 2026/2027

Made by : Mariam H., Claire D. and Azzedine A.

Cervical cancer: preventable, yet unequal

662,044 new cases in 2022

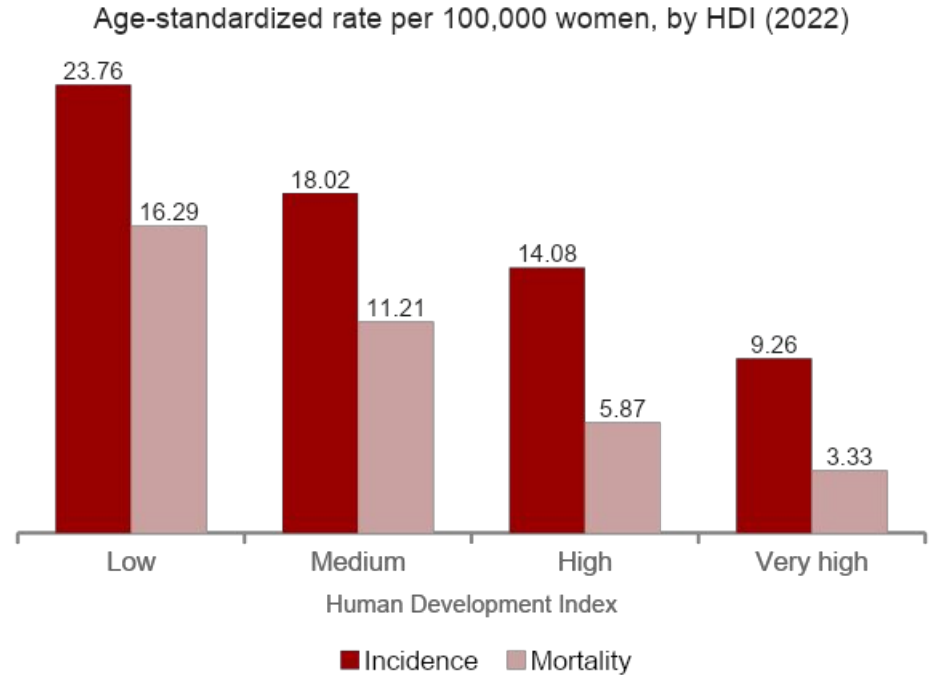
348,709 deaths in 2022

4th most common cancer in women

Necessary cause: persistent high-risk HPV (99.7% of cases)

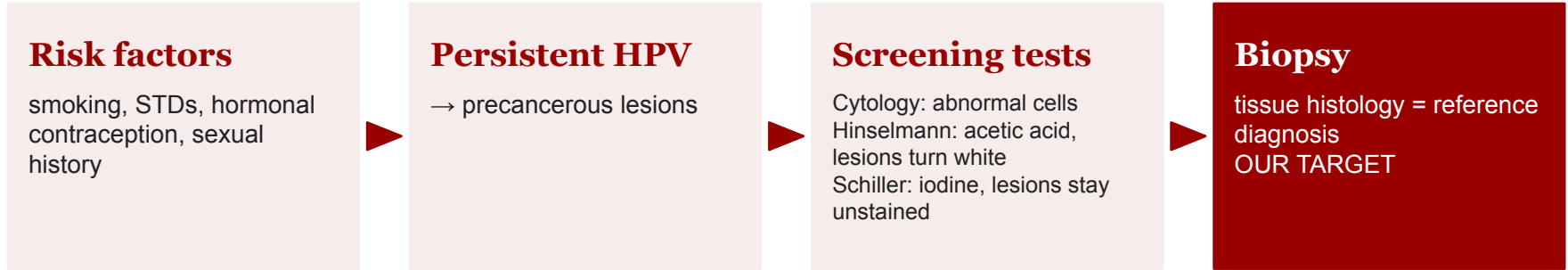
Cofactors: smoking, long-term hormonal contraception, STDs, early first intercourse

WHO 2020: elimination strategy, 90–70–90 targets by 2030



Data: GLOBOCAN 2022 (Wu et al., 2025; Bray et al., 2024). Chart made by the team.

Two levels of information, one question



Scenario 1: risk factors only

Scenario 2: risk factors + screening tests

*Can biopsy outcome be predicted from risk factors alone,
or is performance driven by screening information?*

Dataset and gap in the literature

858

patients (Caracas,
Venezuela)

36

variables: demographics,
habits, medical history, 4
tests

55

positive biopsies = 6.4%
(target)

89%

of positive biopsies had ≥ 1
positive screening test

Prior work on this dataset

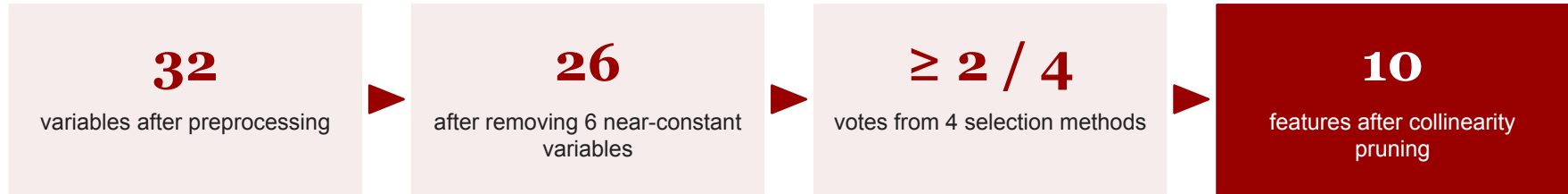
- Tanimu et al., 2022; Shakil et al., 2024: very high accuracy
- But screening tests are used as predictors
- And accuracy is misleading at 6.4% prevalence: always predicting "negative" = 93.6%

Our approach

- Two scenarios: risk factors only vs. + screening tests
- PR-AUC as main metric (focus on the rare positive class)
- Nested cross-validation + held-out test set

Fernandes et al., 2017 · UCI Machine Learning Repository / Kaggle

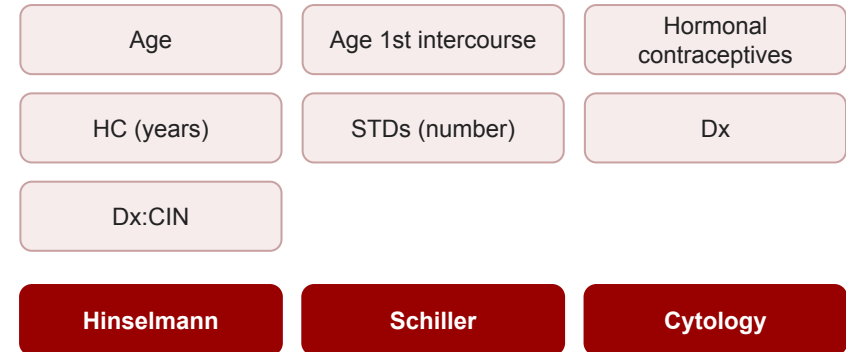
Feature selection: from 32 variables to 10



Four complementary methods

	Classical	Non-linear
Filter	Fisher exact / Mann–Whitney tests	Mutual information
Embedded	L1 (Lasso) logistic regression	RF permutation importance

Selected features



Imbalance: class weighting vs SMOTE

■ risk factors (scenarios 1 & 2) ■ screening tests (scenario 2 only)

From a broad benchmark to two complementary models

Models tested :
Logistic Regression,
Decision Tree, Random
Forest, SVM, XGBoost

Benchmark :
Model comparison

Deeper analysis

Hyperparameters selected by inner CV
Performance estimated by outer-repeated CV (5 x 5)
Primary metric : PR-AUC (Average Precision)

Logistic Regression :

- Pipeline = StandardScaler
- class_weight = 'balanced'
- solver = 'liblinear'
- max_iter=5000
- regularization = L1 or L2 penalty
- C = [1.e-03, 1.e-02, 1.e-01, 1.e+00, 1.e+01, 1.e+02]

Random Forest :

- n_estimators = 500
- bootstrap = 'True'
- class_weight = 'balanced'
- criterion = 'gini'
- max_depth = [None, 3, 5, 8, 12]
- min_samples_leaf = [1, 2, 4, 8, 12]
- max_features = ['sqrt', 0.5, 0.75, 1.0]

⇒ LR and RF represent two complementary modelling assumptions : linear/
interpretable vs non linear / interaction aware

Do we need model complexity to recover additional predictive signal ?

Model 1 : Logistic Regression

- Linear, regularized and interpretable
- Useful to identify the direction and relative contribution of predictors

Model 2 : Random Forest

- Nonlinear and interaction-aware
- Useful to test whether more complex relationships improve prediction

⇒ Two main hypotheses

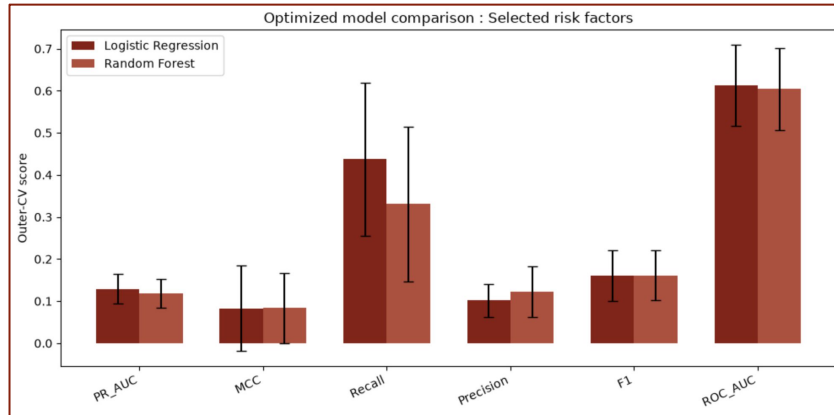
- **Hypothesis 1** : If Random Forest strongly outperforms Logistic Regression in the risk-factor-only scenario
 - Useful information may be hidden in nonlinear effects or interactions
- **Hypothesis 2** : If both models remain weak with risk factors alone but improve with screening tests
 - Predictive performance is mainly limited by the information contained in the predictors

Diagnostic information, not model complexity, drives performance

Scenario 1 : Risk factors only

→ Weak predictive signal

- ◆ $\text{PR-AUC}_{\text{LogisticRegression}} = 0.138$
- ◆ $\text{PR-AUC}_{\text{RandomForest}} = 0.132$

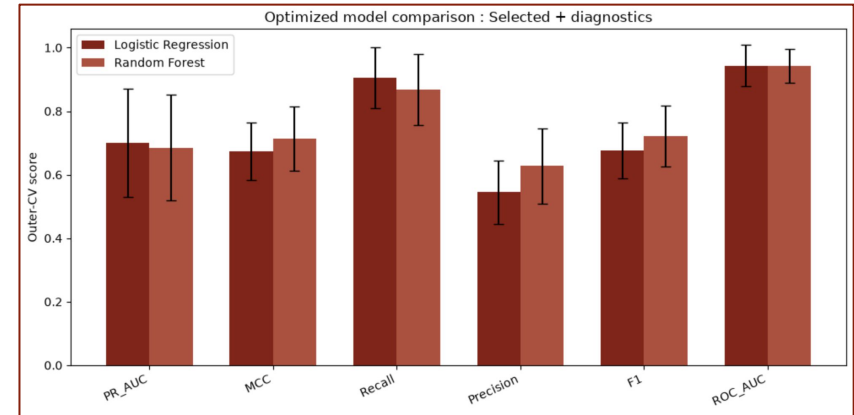


Additional nonlinear complexity does not recover useful signal

Scenario 2 : Screening tests + risk factors

→ Strong improvement

- ◆ $\text{PR-AUC}_{\text{LogisticRegression}} = 0.705$
- ◆ $\text{PR-AUC}_{\text{RandomForest}} = 0.704$



Both models perform similarly despite very different complexity

Screening information markedly improves biopsy prediction with a simple interpretable model

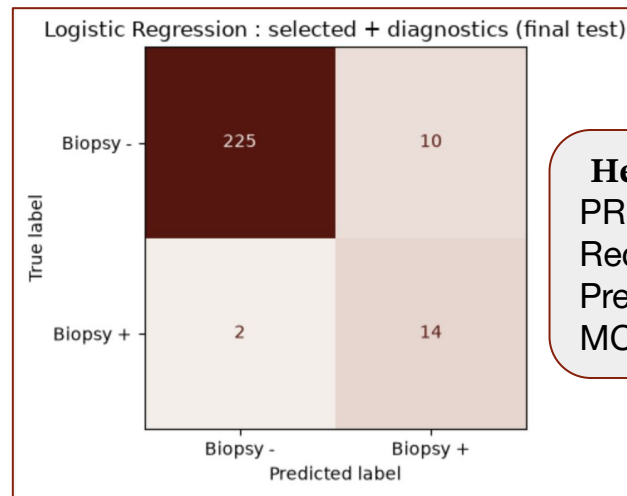
Final model : Screening tests + risk-factors Logistic Regression

- `class_weight = 'balanced'`
- `regularization = L2`
- `Decision threshold = 0.61`

Why Logistic Regression ?

- Similar predictive performance to RF
 - Simpler
 - More interpretable
 - Slightly higher PR-AUC

⇒ Main gain comes from biologically relevant information, not from increasing algorithmic complexity



Held-out test set :

PR-AUC = 0.720
Recall = 0.875
Precision = 0.583
MCC = 0.692

**14 out of 16 positive biopsies
detected**
Only 2 false negatives

Literature Review: Risk Factors Can Influence Cancer Development

Age

→ Cervical Cancer incidence increases with age: onset in 30s → detection ≥50 years old

Age 1st intercourse

→ Greater time/risk for HPV infections and persistence

Dx/CIN

→ Cervical Intraepithelial Neoplasia (CIN I-III): LSIL + HSIL

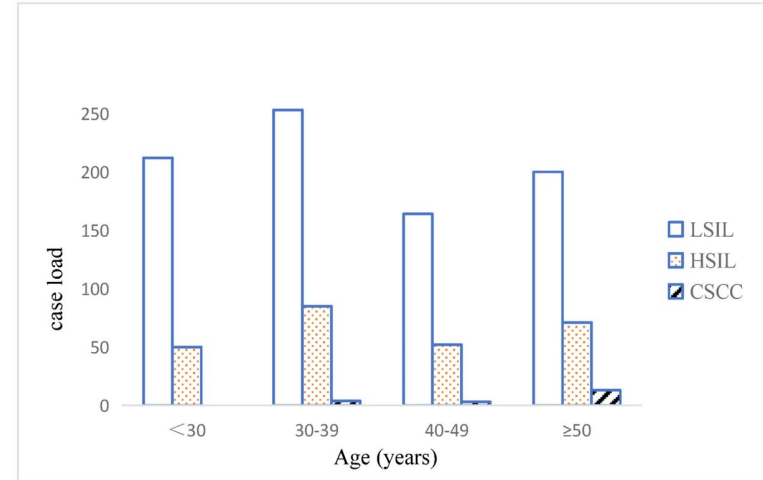
HC/years

→ Risk increases with oral contraceptives, decreases upon cessation

STDs #

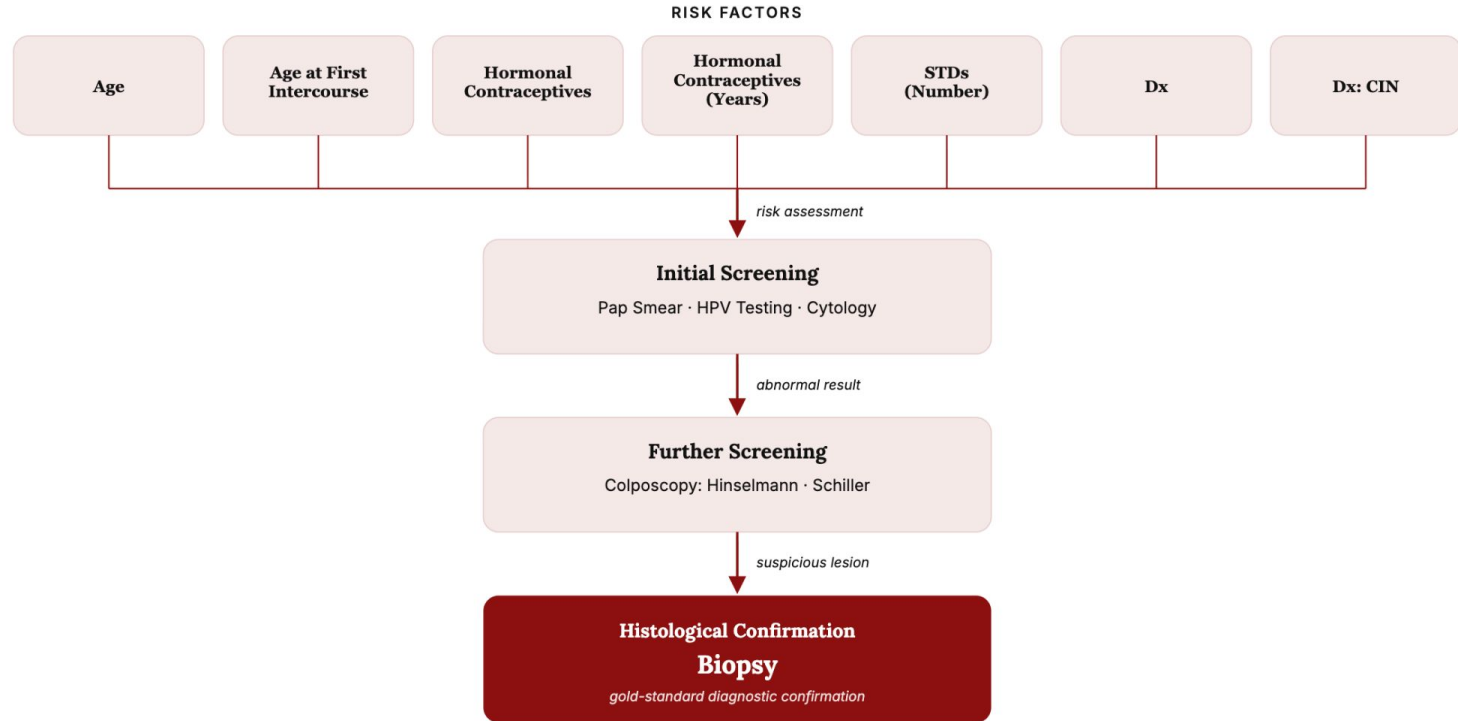
→ Most HPV infections clear naturally, persistence linked to non-HPV STIs as cofactors

(Hu et al., 2026)



LSIL (CIN I/II): Low-grade squamous intraepithelial lesions
HSIL (CIN III): High-grade squamous intraepithelial lesions

Our Model: Risk Factors Not Enough, Cytology + Colposcopy Tests Needed



Difficulties / Limitations

- **Strong class imbalance**
 - Performance estimates remain sensitive to sampling variability
- **Missing-value assumptions**
 - Missing binary exposures were imputed as zero, potentially conflating unknown exposure with true absence
- **Diagnostic proximity to the outcome**
 - Schiller, Hinselmann and Citology are much closer to biopsy outcome than upstream epidemiological risk factor, strong diagnostic-inclusive performance must not be interpreted as equally strong prediction from risk factors alone
- **Predictive rather than causal interpretation**
 - Logistic Regression coefficients and Random Forest permutation importance describe model behaviour, not causal biological effects
- **Statistical rather than clinical threshold**
 - The selected probability thresholds maximize MCC on training OOF predictions, they are not clinically validated operating thresholds
- **No external validation**
 - The held-out test set provides an internal hold-out evaluation only, independent external validation would be required before considering clinical application

Prevention/Diagnostic Measures in LMICs To Prevent Cervical Cancer

Can biopsy outcome be predicted from risk factors alone, or is performance driven by screening information?

Risk Factors: Little Prediction, Difficult to Manage at Population Level (Personal Autonomy)



Cytology/Colposcopy Tests: More Predictive, More Useful to Target/Treat Cancer

Limitations:

- In low/middle-income countries(LMICs):
 - Highest disease burden
 - Colposcopy tests remain scarce
- Accuracy of colposcopy dependent on training/expertise of colposcopists
 - Not easily attainable in LMICs

Future Directions:

- HPV-DNA test most cost-effective vs Pap smear cytology in LMICs
 - Screening at 35, every 5 years, doing test 2-3 times in life = greater success
- Other effective CC prevention strategy: HPV vaccination

References

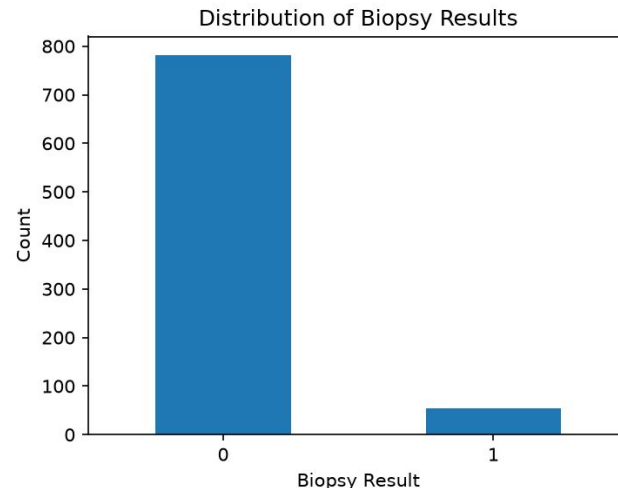
- [1] Bray F. et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin*, 2024;74(3):229–263.
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- [3] Walboomers J.M. et al. Human papillomavirus is a necessary cause of invasive cervical cancer worldwide. *J Pathol*, 1999;189(1):12–19.
- [4] International Collaboration of Epidemiological Studies of Cervical Cancer. Cervical cancer and hormonal contraceptives. *Lancet*, 2007;370:1609–1621.
- [5] World Health Organization. Global strategy to accelerate the elimination of cervical cancer as a public health problem, 2020.
- [6] Fernandes K. et al. Transfer learning with partial observability applied to cervical cancer screening. *Pattern Recognition and Image Analysis*, Springer, 2017, 243–250.
- [7] Tanimu J.J. et al. A machine learning method for classification of cervical cancer. *Electronics*, 2022;11(3):463.
- [8] Shakil R. et al. A precise machine learning model: detecting cervical cancer using feature selection and explainable AI. *J Pathol Inform*, 2024;15:100398.
- [9] Hu, D., Zhao, Y., Han, M. et al. Analysis of factors associated with cervical cancer and precancerous lesions. *Sci. Rep.* **16**, 9600 (2026).
- [10] Phang Romero Casas, C., Ribeiro de Albuquerque, R.d.C., Borge Loureiro, R., Gollner, A.M., Gonçalves de Freitas, M., do Nascimento Duque, G.P. & Kodaira Viscondi, J.Y. Cervical cancer screening in low- and middle-income countries: A systematic review of economic evaluation studies. *Clinics (Sao Paulo)* 77, 100080 (2022).

Link to Git repository of this project : https://github.com/mariamhcn/UM5BM753_Python_Project

Supplementary Materials

Class Imbalance/ Feature Selection Methods Reasoning

- Class weighting>SMOTE
 - (class_weight = “balanced”)
 - Creates greater error for false positive biopsy prediction in model
 - SMOTE creates synthetic data – could add noise/ complications in CV – more data leakage
 - Issue for binary values
 - 0.15>0.12 PR-AUC
- Variance Threshold
 - Eliminated 6 variables with <0.01 variance
- Filter Methods
 - Fisher/Mann-Whitney Correlation (linear)
 - Used appropriately for data types (binary/continuous)
 - Mutual Information (non-linear)
 - Used as another method to see correlation values non-linearly



LR PR-AUCs	Weighting	SMOTE
Risk	0.15+/-0.09	0.12+/-0.08
Risk + Diag	0.72+/-0.16	0.63+/-0.16

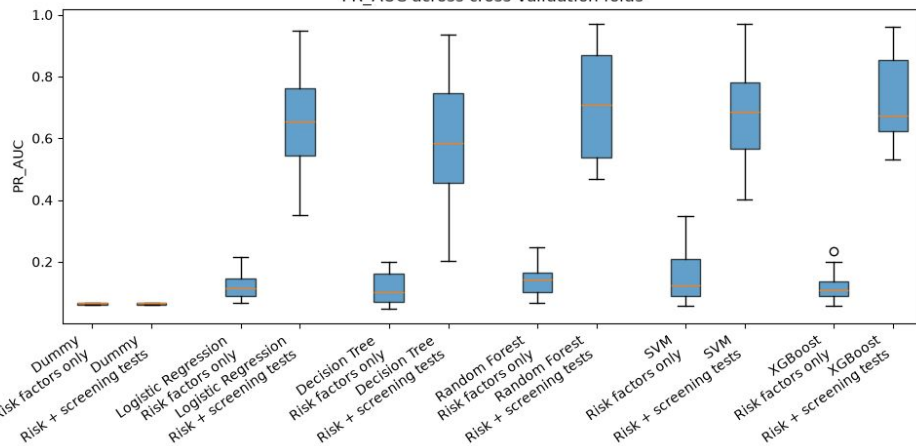
Class Imbalance/ Feature Selection Methods Reasoning

- No Wrapper/RFE CV
 - Unstable across runs depending on if positive biopsies included (hard with so little)
 - More computationally expensive
- Embedded Methods
 - Produce continuous signal (coefficient magnitude) refit over many folds then counted- steadier
 - Lasso/L1 (LogReg - linear)
 - RF permutation importance (non-linear)
- Dropped lower MI variables from colinearity
- Selected features had to pass 2+/4 tests to be included
- Features:
 - Age, Hinselmann, Schiller, Cytology, Dx, Dx:CIN, Hormonal Contraceptives, Hormonal Contraceptives (years), Age 1st sexual intercourse, STDs (number)

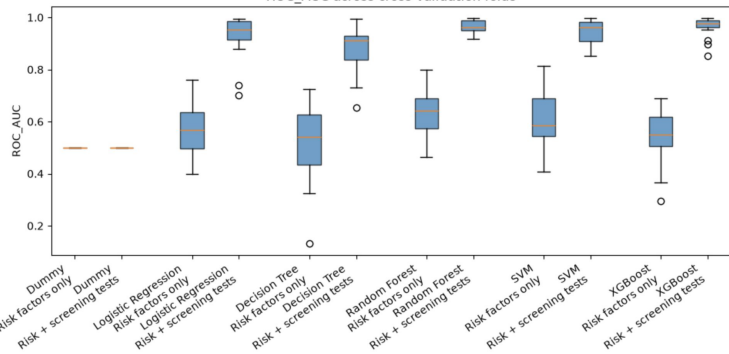
	Classical	Non-linear
Filter	Fisher exact / Mann–Whitney tests	Mutual information
Embedded	L1 (Lasso) logistic regression	RF permutation importance

Benchmark metrics : Why LR and RF selected ?

PR_AUC across cross-validation folds



ROC_AUC across cross-validation folds



Scenario 2 : screening tests + risk

	Model	PR-AUC	MCC	Recall
0	Dummy	0.065 ± 0.004	0.0 ± 0.0	0.0 ± 0.0
1	Logistic Regression	0.656 ± 0.167	0.678 ± 0.11	0.833 ± 0.123
2	Decision Tree	0.606 ± 0.186	0.61 ± 0.122	0.804 ± 0.164
3	Random Forest	0.703 ± 0.172	0.74 ± 0.095	0.851 ± 0.123
4	SVM	0.683 ± 0.145	0.653 ± 0.124	0.781 ± 0.152
5	XGBoost	0.729 ± 0.13	0.714 ± 0.106	0.836 ± 0.134

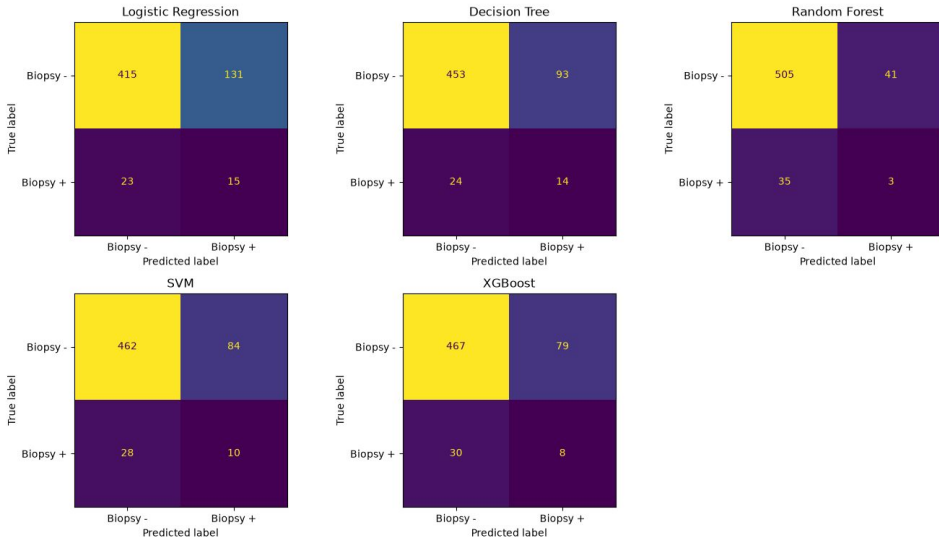
Scenario 1 : risk factors

	Model	PR-AUC	MCC	Recall
0	Dummy	0.065 ± 0.004	0.0 ± 0.0	0.0 ± 0.0
1	Logistic Regression	0.121 ± 0.037	0.07 ± 0.066	0.377 ± 0.129
2	Decision Tree	0.112 ± 0.048	0.095 ± 0.134	0.297 ± 0.179
3	Random Forest	0.138 ± 0.048	0.022 ± 0.106	0.084 ± 0.097
4	SVM	0.15 ± 0.082	0.07 ± 0.109	0.287 ± 0.144
5	XGBoost	0.118 ± 0.043	0.029 ± 0.062	0.189 ± 0.094

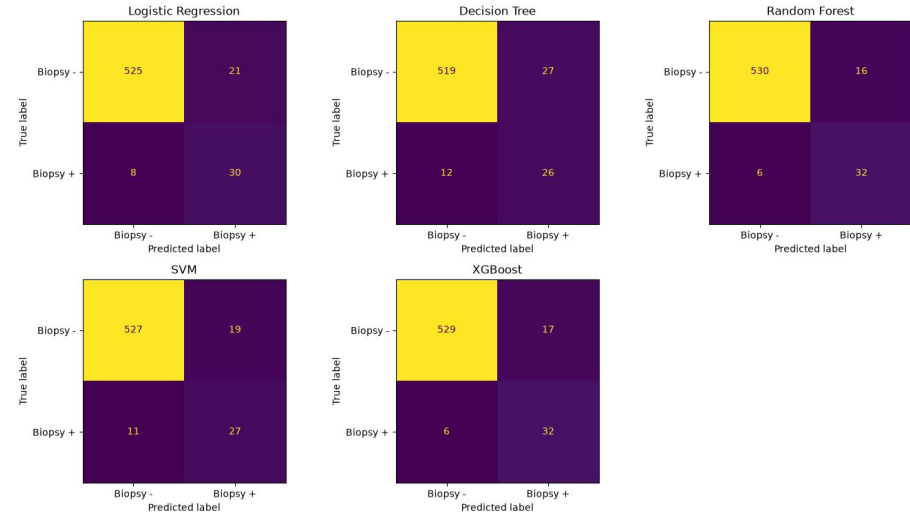
NB : We used a Dummy classifier as a baseline to verify that the real models were learning useful predictive signal beyond the class distribution.

Benchmark confusion matrices : Why LR and RF selected ?

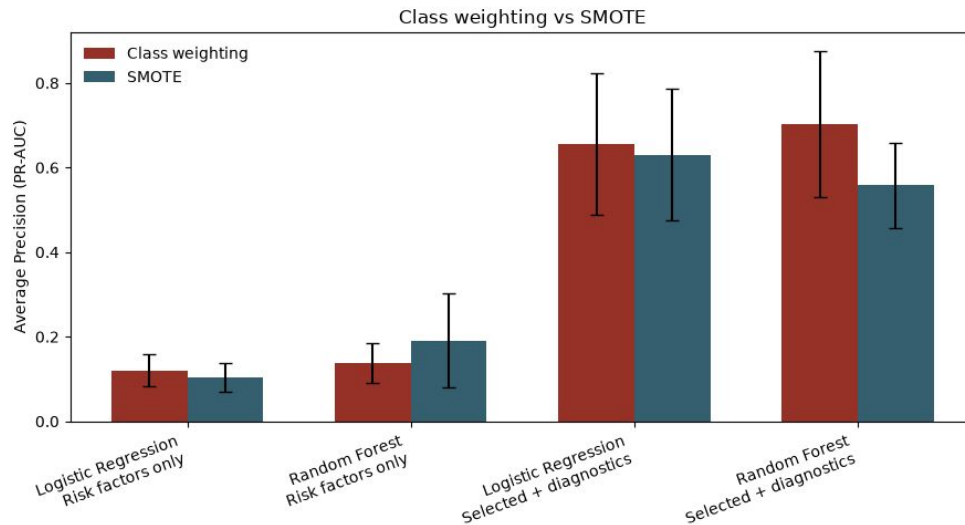
Cross-validated confusion matrices — Selected risk factors



Cross-validated confusion matrices — Selected features + diagnostic tests

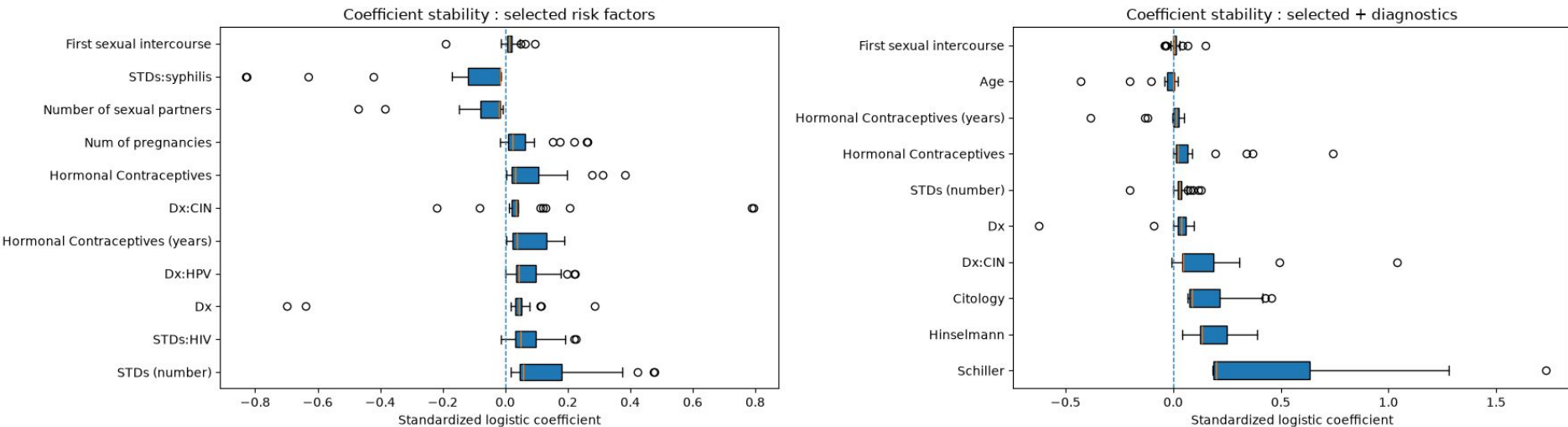


Additional analysis : comparaison of class-weighting and SMOTE



	Scenario	Algorithm	Strategy	PR_AUC_mean	Recall_mean	Precision_mean	F1_mean	Balanced_accuracy_mean	MCC_mean	ROC_AUC_mean
0	Risk factors only	Logistic Regression	Class weighting	0.121	0.377	0.095	0.150	0.561	0.070	0.572
1	Risk factors only	Logistic Regression	SMOTE	0.103	0.401	0.070	0.119	0.516	0.016	0.527
2	Risk factors only	Random Forest	Class weighting	0.138	0.084	0.089	0.083	0.510	0.022	0.630
3	Risk factors only	Random Forest	SMOTE	0.191	0.104	0.189	0.128	0.534	0.091	0.645
4	Selected + diagnostics	Logistic Regression	Class weighting	0.656	0.833	0.597	0.690	0.896	0.678	0.933
5	Selected + diagnostics	Logistic Regression	SMOTE	0.631	0.826	0.616	0.700	0.894	0.687	0.933
6	Selected + diagnostics	Random Forest	Class weighting	0.703	0.851	0.680	0.751	0.911	0.740	0.967
7	Selected + diagnostics	Random Forest	SMOTE	0.559	0.773	0.566	0.648	0.865	0.632	0.952

Logistic regression : Coefficient stability

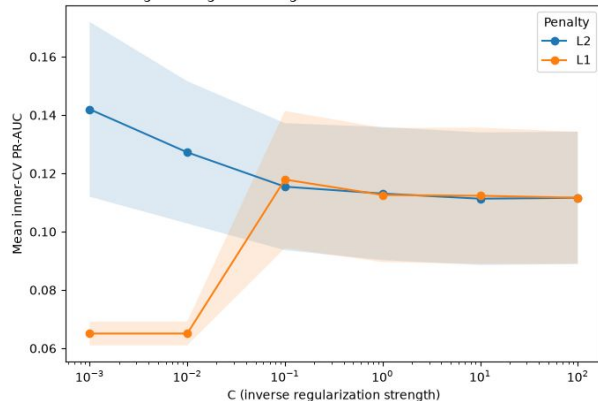


Screening variables show positive coefficients across folds. Whereas, some clinical and missingness variables exhibit greater instability. This supports the predictive importance of diagnostic tests while highlighting uncertainty for weaker predictors.

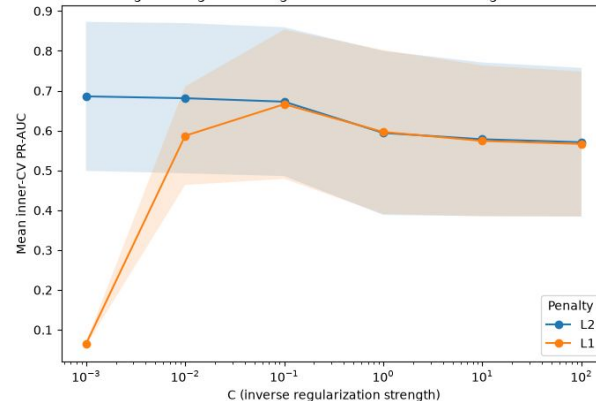
Risk factors – threshold maximizing MCC		Selected + diagnostics – threshold maximizing MCC	
Threshold	0.510	Threshold	0.610
Precision	0.120	Precision	0.615
Recall	0.368	Recall	0.842
F1	0.181	F1	0.711
Balanced_accuracy	0.590	Balanced_accuracy	0.903
MCC	0.111	MCC	0.698

Logistic regression : Hyperparameters selection

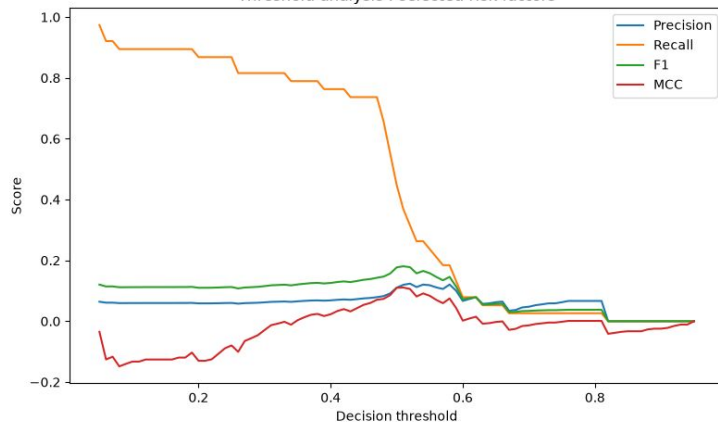
Logistic Regression regularization : selected risk factors



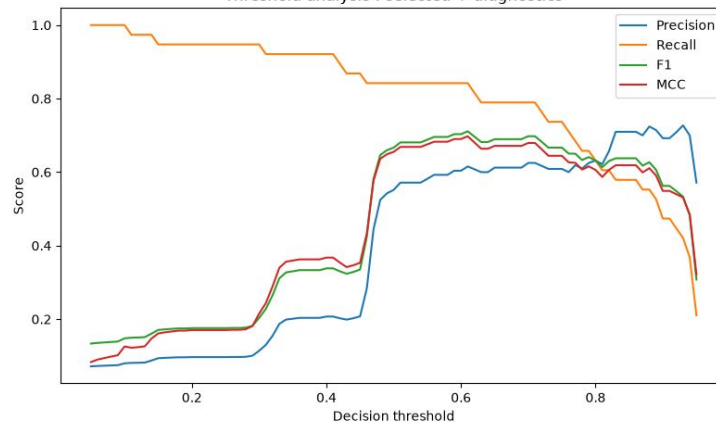
Logistic Regression regularization : selected + diagnostics



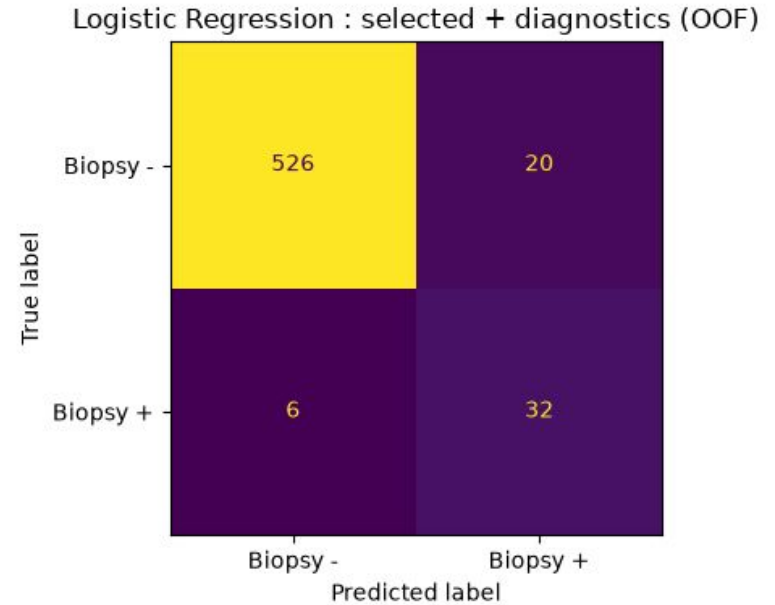
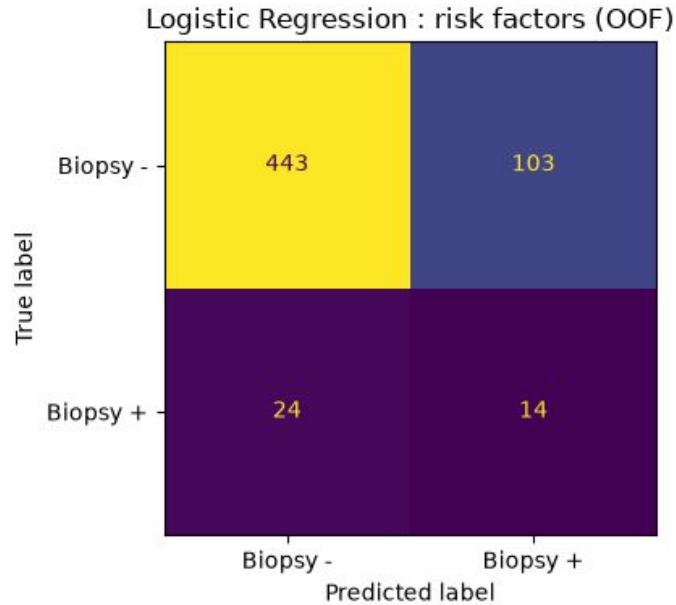
Threshold analysis : selected risk factors



Threshold analysis : selected + diagnostics

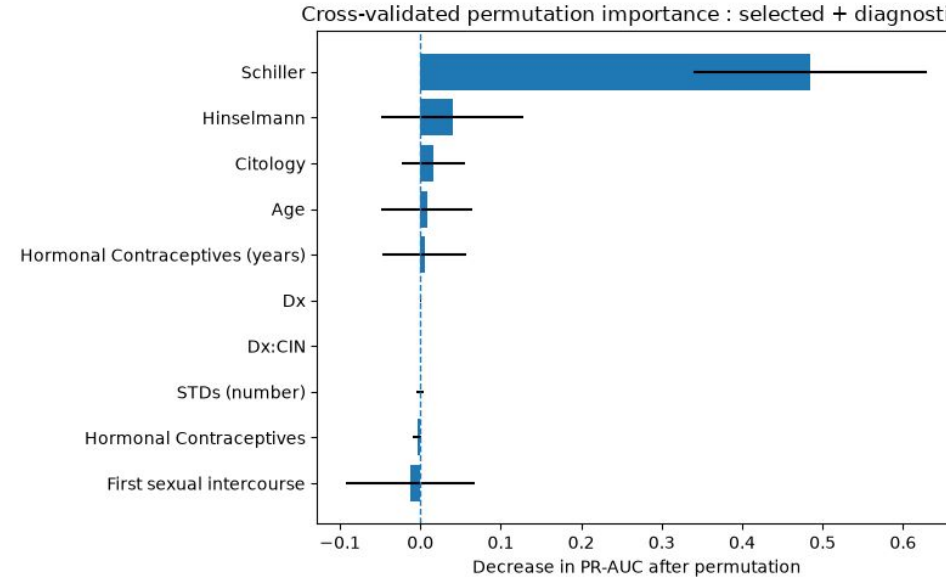
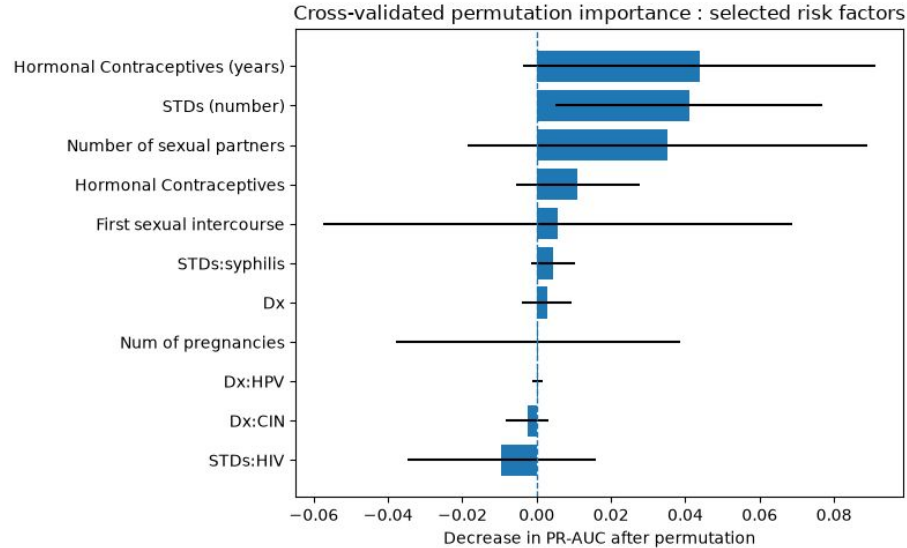


Logistic regression : OOF matrices



These out-of-fold predictions provide an internal estimate of classification behaviour, but the held-out test set is required for the final unbiased evaluation.

Random Forest

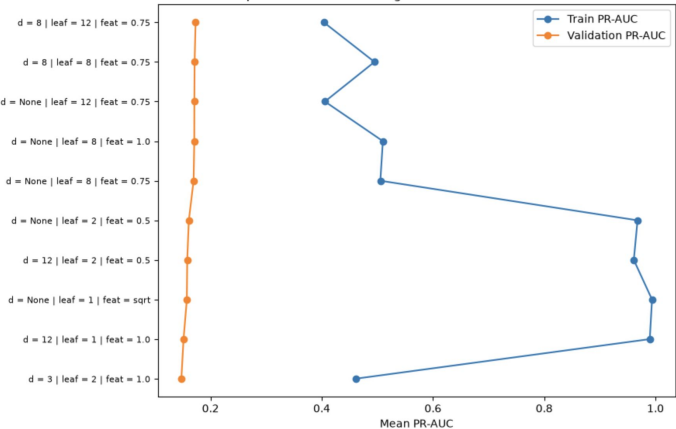


Permutation importance shows that Random Forest predictions are largely driven by the Schiller test, while the remaining predictors provide substantially smaller and less stable contributions across validation folds.

Negative values can occur because of sampling variability (especially as the dataset is small) and shouldn't be interpreted as evidence that a variable is biologically protective.

Random Forest : Hyperparameters selection

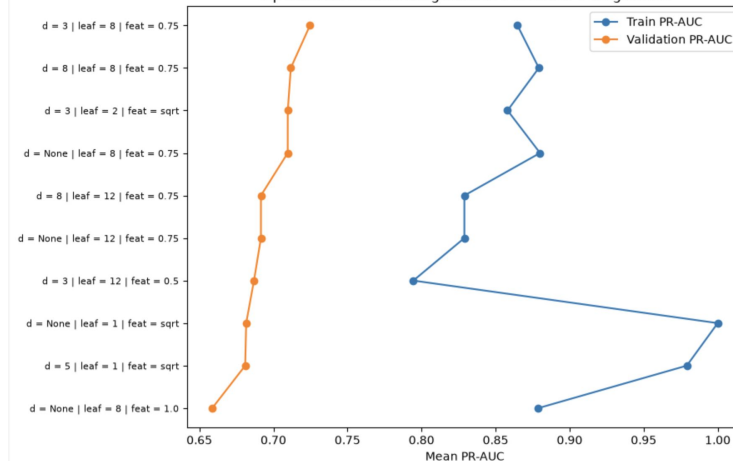
Top Random Forest configurations : selected risk factors



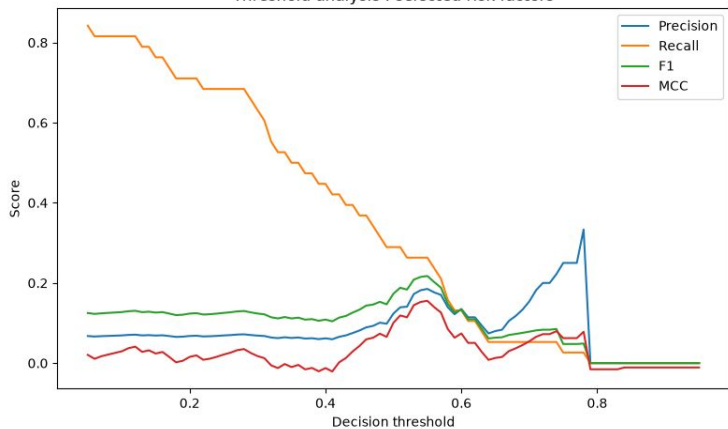
Risk factors : threshold maximizing MCC

Threshold	0.550
Precision	0.185
Recall	0.263
F1	0.217
Balanced_accuracy	0.591
MCC	0.155
Selected + diagnostics : threshold maximizing MCC	
Threshold	0.450
Precision	0.647
Recall	0.868
F1	0.742
Balanced_accuracy	0.918
MCC	0.730

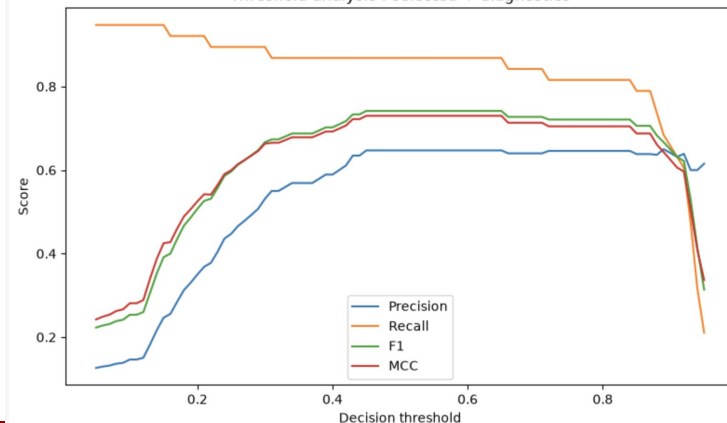
Top Random Forest configurations : selected + diagnostics



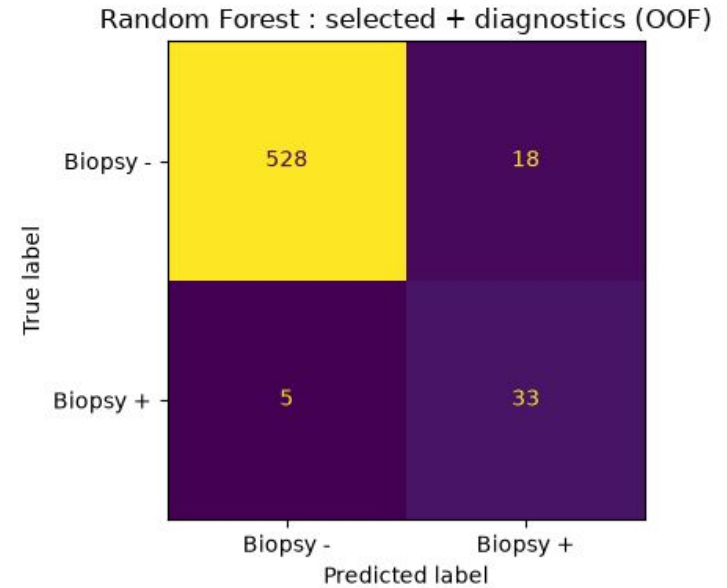
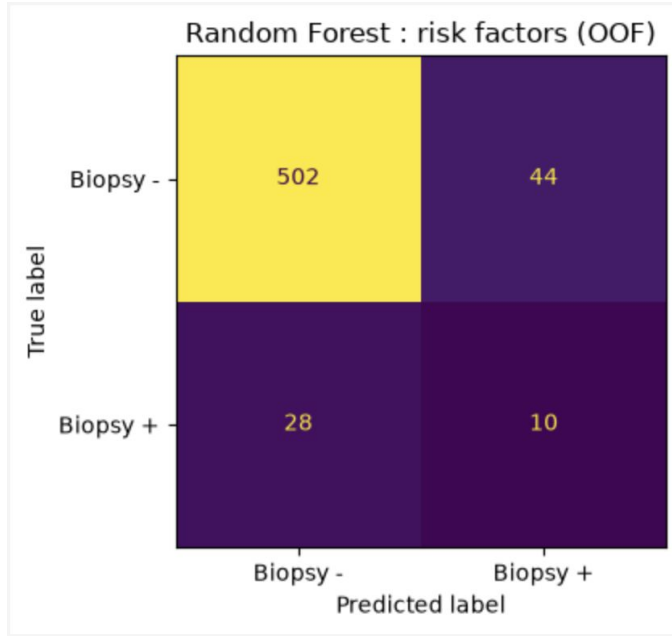
Threshold analysis : selected risk factors



Threshold analysis : selected + diagnostics



Random Forest : OOF matrices



The confusion matrices below are generated from out-of-fold training predictions using the threshold selected from the training data.

These out-of-fold predictions provide an internal estimate of classification behaviour, but the held-out test set is required for the final unbiased evaluation.

Final comparison

Direct comparison of Logistic Regression and Random Forest

Mean performance and variability across the outer folds are compared side by side.

Selected risk factors

Model	Mean		Std	
	Logistic Regression	Random Forest	Logistic Regression	Random Forest
Metric				
Balanced_accuracy	0.591	0.551	0.067	0.075
F1	0.173	0.137	0.058	0.093
MCC	0.103	0.065	0.080	0.105
PR_AUC	0.138	0.132	0.041	0.051
Precision	0.109	0.100	0.042	0.073
ROC_AUC	0.598	0.560	0.091	0.114
Recall	0.462	0.256	0.139	0.183

Selected + diagnostics

Model	Mean		Std	
	Logistic Regression	Random Forest	Logistic Regression	Random Forest
Metric				
Balanced_accuracy	0.923	0.912	0.044	0.058
F1	0.660	0.685	0.083	0.110
MCC	0.661	0.681	0.082	0.112
PR_AUC	0.705	0.704	0.175	0.162
Precision	0.527	0.580	0.096	0.142
ROC_AUC	0.943	0.960	0.056	0.050
Recall	0.906	0.874	0.087	0.116

Final parameters

Logistic Regression:

```
{'risk_factors': {'classifier__C': 0.001, 'classifier__l1_ratio': 0.0}, 'selected_plus_diagnostics': {'classifier__C': 0.001, 'classifier__l1_ratio': 0.0}}
```

Random Forest:

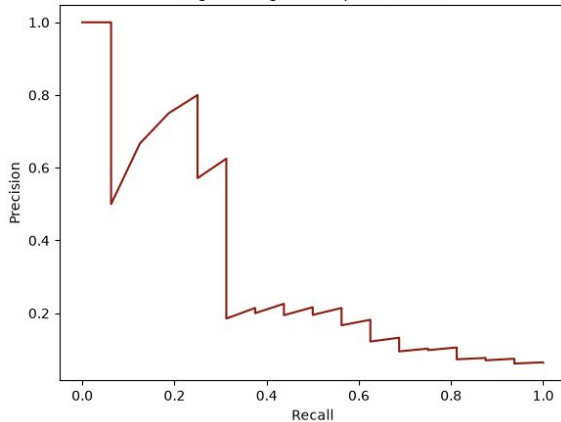
```
{'risk_factors': {'min_samples_leaf': 12, 'max_features': 0.75, 'max_depth': 8}, 'selected_plus_diagnostics': {'min_samples_leaf': 8, 'max_features': 0.75, 'max_depth': 3}}
```

Final held-out test results

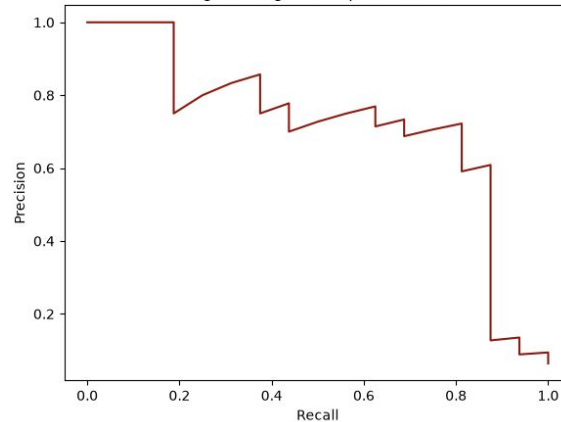
	Scenario	Model	Threshold	PR_AUC	ROC_AUC	Recall	Precision	F1	Balanced_accuracy	MCC
0	Selected risk factors	Logistic Regression	0.51	0.341	0.727	0.500	0.195	0.281	0.680	0.238
1	Selected + diagnostics	Logistic Regression	0.61	0.720	0.924	0.875	0.583	0.700	0.916	0.692

Final comparison - Test

Final PR curve : risk factors
Logistic Regression | AP = 0.341



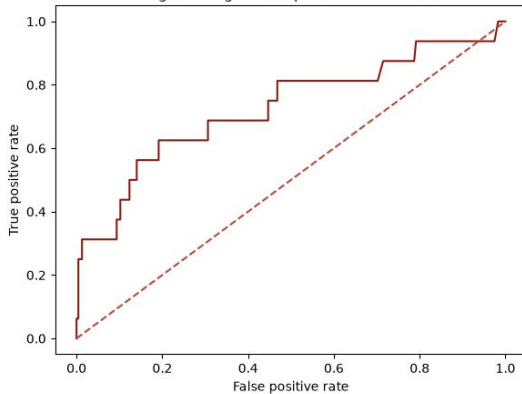
Final PR curve : selected + diagnostics
Logistic Regression | AP = 0.720



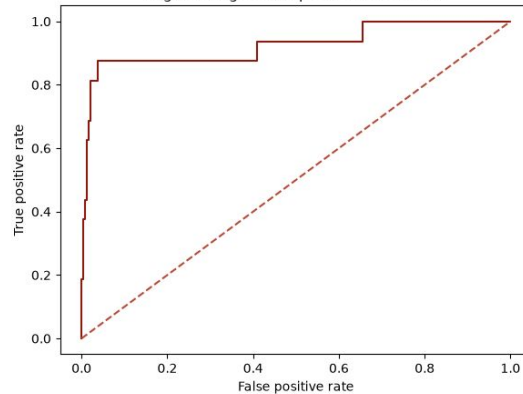
Logistic Regression : risk factors (final test)

True label	Predicted label	
	Biopsy -	Biopsy +
Biopsy -	202	33
Biopsy +	8	8

Final ROC curve : risk factors
Logistic Regression | ROC-AUC = 0.727



Final ROC curve : selected + diagnostics
Logistic Regression | ROC-AUC = 0.924



Logistic Regression : selected + diagnostics (final test)

True label	Predicted label	
	Biopsy -	Biopsy +
Biopsy -	225	10
Biopsy +	2	14